University Management Network Project

استمارة تقييم مشروع مقرر شبكات الحاسب

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عنوان المشروع	(University Management)
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بيانات الطلاب

م	اسم الطالب	الرقم الجامعي	دور الطالب في المشروع
1	Shereen Ayman	24030302	Network Design and Topology, IP Addressing and Subnetting
2	menna wasem	24030342	VLAN Configuration, Inter-VLAN Routing, Switching and Security
3	Omnia Muhammed	24030099	Routing Protocols, NAT/PAT, ACL Configuration
4	Menna Mahmoud	24030153	DHCP, DNS, VPN, QoS, VoIP, Testing and Documentation

تقييم المشروع

معيار التقييم	الدرجة	الملاحظات
ارتباط المشروع بالمقرر وأهدافه	/3	
التنفيذ العملي والجانب التقني	/5	
جودة التصميم والتنظيم والتوثيق	/3	
الابتكار والإبداع	/2	
العرض التقديمي والمناقشة	/4	
مشاركة الطالب والتعاون داخل الفريق	/3	

مجموع درجات المشروع	/20
التقدير النهائي	---

ملاحظات اللجنة

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الإسم	التوقيع
عضو هيئة التدريس	---
عضو الهيئة المعاونة	---

Project Overview

This project represents a complete network design for a University Management System using Cisco Packet Tracer.

The network was designed to connect all university departments securely and efficiently, including Administration, Student Affairs, Faculty Offices, Computer Labs, Library, and Servers.

The project aims to provide fast communication, secure data transfer, and reliable network services inside the university environment.

1. Subnetting

Subnetting was implemented by dividing the university network into smaller subnets for each department.

Implementation:

Each department was assigned a separate IP range to reduce congestion and improve network organization.

Example:

Administration Department → 192.168.10.0/24

Student Affairs → 192.168.20.0/24

Computer Labs → 192.168.30.0/24

Library → 192.168.40.0/24

Benefits:

Better network management

Reduced broadcast traffic

Improved performance

2. VLANs (Virtual Local Area Networks)

VLANs were created to separate university departments logically on the switches.

Implementation:

VLAN 10 → Administration

VLAN 20 → Student Affairs

VLAN 30 → Labs

VLAN 40 → Library

Each switch port was assigned to its related VLAN.

Benefits:

Increased security

Better traffic control

Easier management

3. Inter-VLAN Routing

Inter-VLAN Routing was configured to allow communication between university departments.

Implementation:

Router-on-a-Stick technique was used by creating subinterfaces for each VLAN on the router.

Purpose:

Allow authorized communication between departments while maintaining separation.

4. Access Control Lists (ACL)

ACLs were configured to control access between departments.

Implementation:

Students were prevented from accessing Administration

servers.

Faculty members were allowed to access shared resources.

Benefits:

Improved network security

Controlled access permissions

5. Routing Protocols

OSPF dynamic routing protocol was implemented between routers.

Implementation:

Routers exchanged routing information automatically using OSPF.

Benefits:

Automatic route updates

Faster communication

Efficient routing

6. NAT / PAT

NAT and PAT were configured to allow university devices to access the Internet.

Implementation:

Private IP addresses inside the university were translated into public IP addresses.

Benefits:

Internet access

IP address conservation

Increased security

7. VPN (Virtual Private Network)

VPN was configured to allow secure remote access to the university network.

Implementation:

Encrypted communication tunnels were created between remote users and the university network.

Benefits:

Secure remote access

Protected data transfer

8. DHCP

DHCP service was configured to assign IP addresses automatically to PCs and devices.

Implementation:

DHCP pools were created for each department.

Benefits:

Simplified network configuration

Reduced manual errors

9. DNS

DNS server was configured to translate domain names into IP addresses.

Example:

university.local → 192.168.50.10

Benefits:

Easier access to university services

User-friendly communication

10. STP (Spanning Tree Protocol)

STP was enabled on switches to prevent switching loops.

Benefits:

Network stability

Prevention of broadcast storms

11. Switch Security

Security features were configured on switches to prevent unauthorized access.

Implementation:

Port Security

MAC Address Limiting

Shutdown of unused ports

Benefits:

Increased network protection

Prevention of unauthorized devices

12. Static and Dynamic Routing

Both static and dynamic routing were implemented.

Static Routing:

Used for manually configured paths.

Dynamic Routing:

Used with OSPF for automatic route exchange.

13. TCP and UDP

Both TCP and UDP protocols were implemented and tested.

TCP:

Used for reliable services such as web browsing and file transfer.

UDP:

Used for fast communication such as VOIP services.

14. VOIP (Voice Over IP)

VOIP technology was implemented to allow voice communication between university departments.

Implementation:

IP Phones were connected through the network using Voice VLANs.

Benefits:

Fast internal communication

Reduced communication cost

15. QoS (Quality of Service)

QoS was configured to prioritize important university traffic.

Implementation:

VOIP traffic was given higher priority than regular data traffic.

Benefits:

Improved call quality

Reduced network delay

Better performance for critical services

Configuration

The project was implemented using Cisco Packet Tracer and Cisco IOS configurations.

The configuration includes:

Router configuration

Switch configuration

VLAN setup

Inter-VLAN Routing

ACL configuration

DHCP and DNS setup

NAT/PAT configuration

VPN setup

Security implementation

Conclusion

This University Management Network Project demonstrates a secure and scalable university network infrastructure using modern networking technologies.

The project improves communication, security, and management between all university departments while providing reliable network services for students, faculty members, and administrators.